

Lab Assignment 06



Inspiring Excellence

Course Code:	CSE111
Course Title:	Programming Language II
Topic:	Inheritance
Number of Tasks:	12 (Classwork: 06, Homework: 06)

[Submit all the Coding Tasks (Homework: Task 1 to 5) in the Google Form shared on buX before the next lab.]

[You are not allowed to change the driver codes of any of the tasks]

CLASSWORK

Task 1

Complete the class **Circle** so that the desired outputs are generated properly.

Given Code	Expected Output
<pre>public class shapeTester { public static void main(String[] args) { Shape s = new Shape(); s.name = "Mobius Strip"; s.color = "Blue"; s.displayInfo(); System.out.println("1====="); Circle c = new Circle(); System.out.println("2====="); c.name = "Circle"; c.color = "Red"; c.radius = 5; c.displayInfo(); System.out.println("3====="); c.area(); } } public class Shape { public String name; public String color; public void displayInfo() { System.out.printf("Name: %s\nColor: %s\n", name, color); } } public class Circle extends Shape { //Your Code Here }</pre>	<pre>Name: Mobius Strip Color: Blue 1===== 2===== Name: Circle Color: Red 3===== Area of Red Circle: 78.54</pre>

Task 2

Given the following classes, write the code for the **BBAStudent** class so that the following output is printed when we run the TestStudent class.

Driver Code and Parent Class	Output
<pre>public class TestStudent{ public static void main(String [] args){ BBAStudent b1 = new BBAStudent(); BBAStudent b2 = new BBAStudent("Humty Dumty"); BBAStudent b3 = new BBAStudent("Little Bo Peep"); b1.details(); System.out.println("1-----"); b2.details(); System.out.println("2-----"); b3.details(); } } public class Student{ private String name = "Just a Student"; private String department = "nothing"; public void setDepartment(String dpt){ this.department = dpt; } public void setName(String name){ this.name = name; } public void details(){ System.out.println("Name : " + name + " Department: " + department); } }</pre>	<pre>Name: Default Department: BBA 1----- Name: Humty Dumty Department: BBA 2----- Name: Little Bo Peep Department: BBA</pre>

Task 3

Given the following classes, write the code for the **Vehicle2010** class to print the following output when we run the Vehicle2010User class.

Driver Code and Parent Class	Output
<pre>public class Vehicle2010User{ public static void main(String[] args){ Vehicle2010 car1 = new Vehicle2010(); System.out.println(car1.getPosition()); car1.moveLowerLeft(); System.out.println(car1.getPosition()); Vehicle2010 car2 = new Vehicle2010(); System.out.println(car2.getPosition()); car2.moveUpperRight(); System.out.println(car2.getPosition()); car2.moveLowerRight(); System.out.println(car2.getPosition()); } } public class Vehicle{ public int x; public int y; public void moveUp(){ y = y+1; } public void moveDown(){ y = y-1; } public void moveLeft(){ x = x-1; } public void moveRight(){ x = x+1; } public String getPosition(){ return "("+ x + ","+ y + ")"; } }</pre>	<pre>(0,0) (-1,-1) (0,0) (1,1) (2,0)</pre>

Task 4

Given the following classes, write the code for the **Cricket_Tournament** and the **Tennis_Tournament** classes derived from **Tournament** so that the following output is generated.

Given Code	Expected Output
<pre>public class TournamentTester { public static void main(String[] args) { Cricket_Tournament ct1 = new Cricket_Tournament(); System.out.println(ct1.info()); System.out.println("-----"); Cricket_Tournament ct2 = new Cricket_Tournament("IPL", 10, "t20"); System.out.println(ct2.info()); System.out.println("-----"); Tennis_Tournament tt = new Tennis_Tournament("Roland Garros", 128); System.out.println(tt.info()); } } public class Tournament { private String name; public Tournament() { this.name = "Default"; } public Tournament(String name) { this.name = name; } public String getName(){ return "Tournament Name: "+name; } }</pre>	<pre>Cricket Tournament Name: Default Number of Teams: 0 Type: No type ----- Cricket Tournament Name: IPL Number of Teams: 10 Type: t20 ----- Tennis Tournament Name: Roland Garros Number of Players: 128</pre>

Task 5

Design the **Dog** and **Cat** class derived from the **Animal** class with appropriate attributes and properties so that the driver code can generate the output given below.

Driver Code and Parent Class	Output
<pre>public class AnimalTester1 { public static void main(String[] args) { Dog dog = new Dog("Buddy", 5, "Brown", "Bulldog"); Cat cat = new Cat("Kitty", 3, "White", "Persian"); System.out.println("1.====="); System.out.println(dog.dogInfo()); System.out.println("2.====="); System.out.println(cat.catInfo()); System.out.println("3.====="); dog.makeSound(); System.out.println("4.====="); cat.makeSound(); } } public class Animal { public String name; public int age; public String color; public Animal(String name, int age, String color) { this.name = name; this.age = age; this.color = color; } public String info() { return "Name: "+name+"\nAge: "+age+"\nColor: "+color+"\n"; } }</pre>	<pre>1.===== Name: Buddy Age: 5 Color: Brown Breed: Bulldog 2.===== Name: Kitty Age: 3 Color: White Breed: Persian 3.===== Brown color Buddy is barking 4.===== White color Kitty is meowing</pre>

Task 6

1	public class A{
2	public int temp = 4;
3	public int sum = 1;
4	public int y = 2;
5	public void methodA(int m, int n){
6	int x = 0;
7	y = y + m + (temp++);
8	x = x + 2 + n;
9	sum = sum + x + y;
10	System.out.println(x + " " + y+ " " + sum);
11	}
12	}
13	public class B extends A {
14	public int x = 6;
15	public void methodB(int m, int n){
16	int y = 0;
17	y = ++y + this.y;
18	x = this.y + 2 + temp;
19	methodA(x, y);
20	sum = x + y + super.sum;
21	System.out.println(x + " " + y+ " " + sum);
22	}
23	}

<pre> A a1 = new A(); a1.methodA(1, 1); B b1 = new B(); b1.methodB(1, 2); </pre>	x	y	sum

HOMEWORK

Task 1

Complete the class **Cow** so that the desired outputs are generated properly.

Given Code	Expected Output
<pre>public class AnimalTester2 { public static void main(String args[]){ Animal a1 = new Animal(); System.out.println("1-----"); a1.details(); System.out.println("2-----"); Cow c1 = new Cow(); c1.name = "Pammy"; System.out.println("3-----"); System.out.println("Name: " + c1.getName()); c1.details(); System.out.println("4-----"); c1.updateSound("Moo"); System.out.println("5-----"); c1.details(); } } public class Animal{ public int legs = 4; public String sound = "Not defined"; public void details(){ System.out.println("Legs: "+legs); System.out.println("Sound: "+sound); } } public class Cow extends Animal{ //Your Code Here }</pre>	<pre>1----- Legs: 4 Sound: Not defined 2----- The cow says hello! 3----- Name: Pammy Legs: 4 Sound: Not defined 4----- 5----- Legs: 4 Sound: Moo</pre>

Task 2

Design the **CheckingAccount** class derived from the **Account** class with appropriate attributes and properties so that the driver code can generate the output given below.

Driver Code and Parent Class	Output
<pre>public class Account{ public double balance = 0.0; public Account(double balance){ this.balance = balance; } public double showBalance(){ return balance; } } //Tester Class public class TestAccount{ public static void main(String [] args){ System.out.println("Total Checking Accounts: "+CheckingAccount.count); CheckingAccount c1 = new CheckingAccount(); System.out.println("Account Balance: " + c1.showBalance()); CheckingAccount c2 = new CheckingAccount(100.0); System.out.println("Account Balance: " + c2.showBalance()); CheckingAccount c3 = new CheckingAccount(200.0); System.out.println("Account Balance: " + c3.showBalance()); System.out.println("Total Checking Accounts: "+CheckingAccount.count); } }</pre>	<pre>Total Checking Accounts: 0 Account Balance: 0.0 Account Balance: 100.0 Account Balance: 200.0 Total Checking Accounts: 3</pre>

Task 3

Given the following classes, write the code for the **Book** and the **CD** class so that the following output is printed.

Driver Code and Parent Class	Expected Output
<pre>public class Tester6 { public static void main(String[] args) { Book book = new Book(1, "The Alchemist", 500, "97806", "HarperCollins"); System.out.println(book.printDetail()); System.out.println("-----"); CD cd = new CD(2, "Shotto", 300, "Warfaze", 50, "Hard Rock"); System.out.println(cd.printDetail()); } } class Product { private int id; private String title; private int price; public Product(int id, String title, int price) { this.id = id; this.title = title; this.price = price; } public String getIdTitlePrice() { return "ID: " + id + " Title: " + title + " Price: " + price; } }</pre>	<pre>ID: 1 Title: The Alchemist Price: 500 ISBN: 97806 Publisher: HarperCollins ----- ID: 2 Title: Shotto Price: 300 Band: Warfaze Duration: 50 minutes Genre: Hard Rock</pre>

Task 4

Design the **ComplexNumber** class with the necessary property to produce the output from the given driver code.

Driver Code and Parent Class	Output
<pre>public class ComplexNumberTester { public static void main(String[] args) { ComplexNumber cn1 = new ComplexNumber(); System.out.println(cn1.details()); System.out.println("-----"); ComplexNumber cn2 = new ComplexNumber(5.0, 7.0); System.out.println(cn2.details()); } } public class RealNumber { public double realValue; public RealNumber() { this(0.0); } public RealNumber(double realValue) { this.realValue = realValue; } public String getReal(){ return "RealPart: " + realValue; } }</pre>	<pre>RealPart: 1.0 ImaginaryPart: 1.0 ----- RealPart: 5.0 ImaginaryPart: 7.0</pre>

Task 5

Given the following classes, write the code for the **CSEStudent** class derived from **Student** so that the following output is generated.

Given Code	Expected Output
<pre>public class StudentTester{ public static void main (String args[]){ CSEStudent.details(); System.out.println("1-----"); CSEStudent s1 = new CSEStudent("Bob", 23); s1.info(); System.out.println("2-----"); CSEStudent s2 = new CSEStudent("Don", 33); s2.info(); System.out.println("3-----"); s1.addLabBasedCourse("CSE220"); s1.addLabBasedCourse("CSE221"); System.out.println("4-----"); s1.info(); System.out.println("5-----"); CSEStudent.details(); System.out.println("6-----"); s1.addLabBasedCourse("CSE230"); System.out.println("7-----"); s1.info(); System.out.println("8-----"); s2.addLabBasedCourse("CSE110"); s2.info(); } } class Student{ public String name; public int id; public String courses = ""; public Student(String n, int i){ name = n; id = i; } public void info(){ System.out.println("Name: "+name); System.out.println("ID: "+id); System.out.println("Courses: "+courses); } }</pre>	<pre>Total CSE Students: 0 Available Lab Based Courses: CSE110 CSE111 CSE220 CSE221 1----- Name: Bob ID: 23 Courses: 2----- Name: Don ID: 33 Courses: 3----- 4----- Name: Bob ID: 23 Courses: CSE220 CSE221 5----- Total CSE Students: 2 Available Lab Based Courses: CSE110 CSE111 CSE220 CSE221 6----- It is not a lab based course! 7----- Name: Bob ID: 23 Courses: CSE220 CSE221 8----- Name: Don ID: 33 Courses: CSE110</pre>

Task 6

1	public class A{
2	public int m = 14, n = -1, p = 3;
3	public static int temp = 11;
4	public void methodA(int m, int n){
5	int x = 0;
6	p = this.p + m + (this.temp++);
7	x = x + 2 + n;
8	this.n = n + B.x + p;
9	System.out.println(this.m + " " + p+ " " + this.n);
10	}
11	}
12	public class B extends A {
13	public static int x = 6;
14	public void methodB(int x, int y){
15	y = ++y + this.p + n;
16	super.p = x + 2 + temp;
17	methodA(temp, x);
18	B.x = this.x + x + super.temp;
19	System.out.println(B.x + " " + y+ " " + A.temp);
20	}
21	}

<pre> B b1 = new B(); b1.methodB(3, 5); A a1 = new A(); a1.methodA(12, -9); </pre>	x	y	sum

Ungraded Tasks (Optional)

(You don't have to submit the ungraded tasks)

Task 1

Given the following classes, write the code for the **Player** and the **Manager** classes derived from SportsPerson class so that the following output is printed. To calculate the match earnings use the following formula:

1. Player: $(\text{total_goal} * 1000) + (\text{total_match} * 10)$
2. Manager: $\text{match_win} * 1000$

Given Code	Expected Output
<pre>public class PlayerTester { public static void main(String[] args) { Player playerOne = new Player("Al-Nassr", "Ronaldo", "Striker", 25, 32); playerOne.calculateRatio(); playerOne.printDetails(); System.out.println("-----"); Manager managerOne = new Manager("Real Madrid", "Zidane", "Manager", 25); managerOne.printDetails(); } } class SportsPerson { private String team; private String name; public String role; public double earningPerMatch; public SportsPerson(String teamName, String name, String role){ this.team = teamName; this.name = name; this.role = role; this.earningPerMatch = 0; } public String getNameTeam() { return "Name: " + name + ", Team Name: " + team; } }</pre>	<pre>Name: Ronaldo, Team Name: Al-Nassr Team Role: Striker Total Goal: 25, Total Played: 32 Goal Ratio: 0.78 Match Earning: 25320K ----- Name: Zidane, Team Name: Real Madrid Team Role: Manager Total Win: 25 Match Earning: 25000K</pre>